

Assignment 4: Part I

Library Automation: Needs Assessment & System Selection Criteria

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Introduction

Describe the library (Amanda)

The Seattle Public Library (SPL) has offered public library services to the Greater Seattle area for 133 years. Throughout that time, SPL librarians and staff have listened to the technological needs and wants of its various communities to instill a sense of belonging. SPL provides free public spaces for all; connects people to books, information, and other resources; provides programming and utilizes outreach and engagement strategies; collaborates with external organizations to boost community impact; and invites member input on services and programs. In addition to a 12 story, 363,000 square foot Central Library, SPL has a network of 26 other library locations throughout the Seattle Metropolitan area (Seattle Public Library, n.d.-b; Seattle Public Library, n.d.-c).

The population of the City of Seattle is over 755,000 people and the population of the greater Seattle Metropolitan area is over 4 million people, many of whom utilize the digital offerings at SPL. On average, the Central Library hosts over 1.2 million visitors annually and the library branches serve over 3.6 million annual visitors. Over 32,000 people attended programs at SPL's Central Library in 2024. SPL loaned over 5.88 million books and other physical materials in the same period. The King County Library System (KCLS), which includes SPL's central location, reported a record-breaking number of digital books loans in 2024. More than 10 million digital materials were loaned to patrons last year (Seattle Public Library, n.d.-c).

The overarching priorities at SPL are equity and intellectual freedom. A wide array of adult and youth services are offered including books and media; research and reference; workshops and classes; job and career services; literacy and language services; computer and technology access; community and cultural programs; storytimes; homework help; summer reading programs; teen programs and events; youth literacy programs; creative arts and STEAM programs; and youth access to digital

resources. In-depth information about all that SPL has to offer can be found at spl.org (Seattle Public Library, n.d.-b).

Needs Analysis

Current Operations (Amanda) and Current Services (Amanda)

The SPL's 2025 spending budget, for all operations and capital improvements, is \$115.8 million. SPL is integrating various aspects of the 10-year Strategic Plan into the 2025 Operations Plan. Operation area foci include technology innovations, technology sustainability, collection development, skill building for all staff members, and library resiliency. Service area foci include literacy, empowerment and enrichment, program creation, and community belonging through space improvements. To maintain and improve library operations and services, SPL is committed to prioritizing:

- Hours and access by using funds to:
 - o Improve branch programming
 - o Provide security and safety services
- Books and materials by altering funds to:
 - o Enrich special collections
 - o Add decodable readers to children's collection
 - o Reduce and retire old technology formats
- Technology and online services by altering funds to:
 - o Eliminate outdated software (including the planning and implementation of an integrated library system (ILS) replacement)
 - o Hire a Cybersecurity Analyst in response to the 2024 cybersecurity attack
 - o Reduce the number of computer hardware replacements and other IT infrastructure replacements

- Literacy and early learning by using funds to:
 - o Continue enrichment program (successful birth through 5-year activity program)
- Building and facility support by using funds to:
 - o Provide ongoing maintenance
 - o Reinforce masonry at University Branch Library
 - o Purchase electric heat pumps and energy recovery ventilators
 - o Purchase solar panels
 - o Complete ADA improvements
 - o Perform seismic reinforcement (Seattle Public Library, n.d.-a)

The Seattle Public Library (SPL) views automation as a very important tool for providing access to library services and wants to use this tool in every way possible bearing in mind current and future budgetary limitations. In the next decade, SPL plans to use automation to extend, improve, and enhance current services in the areas of reference, circulation, patron self-service, and collection development, as well as to add new automation features in areas such as communications, records management, and acquisitions. SPL will be exploring more modern technologies, comparing cost efficiency measures with other large library systems, and highlighting the measures that will improve access and internal use in a timely, accurate, and comprehensive way. Operations and services at SPL will continue to emphasize lifelong learning, access to information, and community enrichment for all.

User Population and User Needs (Blake)

The city of Seattle has seen sharp population growth in recent years, rising 21.1% between 2010 and 2020, a steep increase against the 14.6% growth observed across Washington State (*About Seattle - OPCD*, n.d.). Of the city's over 700,000 residents, a majority of the population lies between age 18 and 65, with 12.8% of the city's population being sixty-five or over (*U.S. Census Bureau QuickFacts*, n.d.). The

highest concentration of age groups is seen between ages 25 and 34, indicating a population that skews toward young adulthood (*About Seattle - OPCD*, n.d.). A diversity of racial and ethnic backgrounds can be found throughout the city as well as a relatively high number of persons who speak a language other than English at home at 23.1% (*U.S. Census Bureau QuickFacts*, n.d.).

The programming branch of SPL operations seeks to address the variety of needs expressed by these populations. SPL's 2025 Operations Plan allocates a significant amount of resources toward early literacy in the community as well as other programming funds (Seattle Public Library, n.d.). Past programming has involved book readings, homework help sessions, partnerships with arts and literacy organizations around the city, curated exhibitions, and an assortment of workshops held both in-person and virtually (*2023 Q2 The Seattle Public Library Levy Report*, n.d.). Events have also been developed in Spanish to better address the needs of Seattle's multilingual population (*2023 Q2 The Seattle Public Library Levy Report*, n.d.).

In 2024 it was determined that one-third of SPL users depend on library resources to access the internet, whether through circulating devices, on-site computers and Wi-Fi, and portable hotspots, illustrating the still pervasive lack of access to digital technology, commonly known as the "digital divide" (Seattle Public Library, n.d.). Digital items saw a 17% increase in circulation in 2023 (*2023 Statistical & Financial Summaries*, n.d.). Compared with the 14% increase seen in the same category in 2022, it remains clear that digital holdings are steadily rising in popularity among SPL users (*2022 Statistical & Financial Summaries*, n.d.). These statistics demonstrate a need for readily available and current digital technology from the city's public library.

Collection Description (Blake)

SPL hosts a collection of 2.8 million items, 1.8 million of which are physical items (*2023 Statistical & Financial Summaries*, n.d.). Notable formats include books, eBooks, video resources, audio

resources, periodicals, streaming media, Wi-Fi hotspots, databases, and language-learning tools (*Your Collection at The Seattle Public Library*, n.d.). 6 million physical circulations occurred in 2023, with 7.4 million digital, an average of over 1 million circulations per month (*2023 Statistical & Financial Summaries*, n.d.).

Recent changes to library collections include a phased retirement of the CD format, as well as cuts to other disc media (Seattle Public Library, n.d.-a). Funds for replacing public-facing computers were also diminished (Seattle Public Library, n.d.-a). Special collections, however, has seen a budget allocation increase in order to properly host the associated metadata, as well as accommodate a larger collection size (Seattle Public Library, n.d.-a).

Large-scale efforts have been made in a push for collection diversity. A 2022 full audit of SPL's physical and digital collections resulted in findings that the library's collection is one of the most diverse in North America (*Equity Work at the Library*, n.d.). The library's 2024 4Q Levy Report reveals that this audit also resulted in the addition of over 4,000 copies of materials to the collection in both physical and digital formats with the mission of further strengthening its diversity (*Library Levy*, n.d.). SPL has committed to reviewing and acquiring items on an ongoing basis against a variety of factors including budgetary constraints, industry norms, and item circulation and condition, with items local to and concerning the history of Seattle being prioritized (*Library Levy*, n.d.).

Existing Technology Infrastructure (Blake)

As a large public library located in a major city within its state, SPL regularly relies on automation tasks in its operation. SPL currently employs a variety of systems to achieve such tasks as circulation, digital content management, offering discovery tools to inquiring patrons, and the automatic sorting of materials. Current systems read as follows (*Seattle Public Library*, n.d.):

- Current ILS: Horizon (since 2004)

- Former ILS: Dynix
- Discovery Interface: BiblioCore (since 2010)
- Digital Collection Management: CONTENTdm
- Web Content Management: Ingeniux
- Item ID: RFID
- Automated Materials Handling: Tech Logic
- Self-Service Checkout Kiosk System: Liber8

Public-facing technologies are also available and may aid patrons in job-seeking, catalog searching, and general productivity (Seattle Public Library, n.d.). Certain technologies, including computers and digital setups for meetings rooms have faced reduced priority in the library's budget (Seattle Public Library, n.d.). Publicly available technology includes the following (*Computers & Equipment*, n.d.):

- Desktop computers
 - Catalog (unlimited use)
 - Express (30 minutes)
 - Job seeking (2 hours)
- Printer
- Copy machine
- ScanEZ kiosks
- Wireless internet access
 - Circulating Wi-Fi hot spots
- Circulating laptops
- Graphing calculators
- Adaptive equipment

- Video conference meeting rooms

Finally, the library also operates self-service checkout kiosks by the company Biblioteka which are used to perform circulation tasks at all branches within the SPL system.

Strengths and Gaps (Bethany)

SPL has ensured that it provides technologically up-to-date library spaces for all its visitors. All SPL locations provide up-to-date hardware such as printers, copy machines, and fax machines using ScanEz kiosks to the public. All locations also have free Wi-Fi and internet access on their public computers, as well as self-checkout kiosks for faster checkouts. They also have circulating hotspots for the public that provide further internet access to library users (Computers & Equipment, 2018). SPL is also at the forefront of assistive technology and software offering ZoomText screen magnification software, ZoomText on all public computers at all locations, refreshable braille displays, a braille embosser, CCTV magnifiers (video magnification systems), a video phone for hard of hearing patrons to make ASL-interpreted phone calls or to converse directly with other ASL signers, and two height-adjustable computer desks (Accessibility Toolkits, n.d.).

SPL is currently using Horizon as its ILS and has been using it since 2006. This software has been providing the circulation of 12 million print and e-books (Turner, 2016). SPL also uses a host of networks to support library functions including OCLC service such as CONTENTdm, which was added in 2008 to support its special collections, OCLC Cataloging and Metadata Subscription as well as EZproxy which provides analytics (Increase Accessibility to Digitized Special Collections, 2025). SPL also has a host of cloud-based services that are accessible to the public like Overdrive and Libby, PressReader, Access Video, and Kanopy (Digital Resources, 2018). It also migrated to SharePoint Online after a ransomware attack in 2024. This multi-factor authentication has been implemented on staff systems.

All public hardware, accessibility software, and cloud-based services are working well within SPL. In 2023 SPL's technology team upgraded 368 public staff computers, replaced 40 printers, and decreased wasteful desktop computer space by 30%, making room for more laptops, scanning stations, and upgraded print kiosks (The Seattle Public Library LEVY REPORT, 2023). Initiatives like upgrading the library's infrastructure to provide more dependable internet access and the Hotspot circulation program in quarter two of 2024, which had 850 hotspots in the general collection that were checked out 1245 times, are just highlights of the many successful technology initiatives (The Seattle Public Library LEVY REPORT, 2024).

Some gaps and challenges that SPL faces in its existing technology are its outdated ILS and its security system. SPL uses Horizon as its ILS and has been using this software since 2004. Horizon's user interface is dated, and developers are not creating new features for this software anymore. The security system of SPL was breached in 2024 during a ransomware attack. SPL's IT team worked hard to recover and restore access to overdrive, Wi-Fi in the building, public computers and printers, and all other networks. This attack caused a decline in patron printing by 27%, and computer usage by 69% compared to 2023 (The Seattle Public Library LEVY REPORT, 2024). Based on the identified gaps, potential solutions are to work with cybersecurity teams to boost security, as well as upgrading SPL's ILS to a solution that is web-based and a cloud-based service to relieve issues related to managing clients on staff computers.

System Selection Criteria

Functional Requirements (Bethany)

1. Circulation Management

- Allow library staff and self-service kiosks to process check-outs using barcodes or RFID
- Bulk check-in/check-out for efficiency
- Track item status like checked out, available, reserved, lost, damaged

- Restrict check-out based on item type, and overdue status
- Allow patrons to renew borrowed items online, via mobile apps, or at the library
- Set renewal limits
- Send automatic notifications when a held item is ready for pickup

2. Patron Management

- Register new patrons with personal details and contact information
- Maintain borrowing history and preferences for personalized services
- Assign borrowing privileges, renewal limits, and fines
- Send email and text message alerts for due dates, overdue items, and hold pickups
- Notify patrons of upcoming library events or policy changes
 - Allow patrons to customize notification preferences

3. Search and Discovery

- Keyword search across multiple fields
 - Title, author, and subject
- Enable faceted filtering
 - Format, availability, location, and the publication year
- Display item status
 - Checked out, available, and on hold
- Show call number for physical items

4. Cataloging Support

- Support classification schemes
 - Dewey Decimal Classification System (DDC)
- Auto indexing to assign search terms

- Detailed item records, including barcode, RFID, and call numbers
- Track multiple copies, volumes, and editions of books

5. Acquisitions Support

- Enable selection and ordering of new materials
- Support integration with vendor systems
- Log received materials and match them against purchase orders
- Generate invoices and track payments

6. Data and Statistical Reporting

- Generate reports on borrowing trends, and other trends
- Track overdue items and unpaid fines
- Track patron activity and borrowing behavior
- Monitor patrons' registrations and expirations

Technical Specifications (Bethany)

1. Scalability

- Cloud-based deployment
- Offering mobile library application features

2. Security

- Multi-factor authentication (MFA) on all staff computers and software
- Passwords or pin numbers
- Staff ID smart cards
- Fingerprint activation
- Backup & disaster recovery
- Intrusion detection & prevention systems

- Monitoring functions watching for potential breaches
- Alerting capability to notify SPL administrators of suspicious activity
- Prevention measures to block malicious traffic or unauthorized user access

Usability and Accessibility (Ashley)

One very critical factor in the selection of an ILS for the Seattle Public Library is usability.

Ensuring that a system is intuitive, efficient, and accessible for both the library staff and the library patrons is one of the most important goals of any library including SPL. To ensure these goals are met, the chosen system must prioritize the following:

1. A user-friendly interface
 - a. A clear and navigable interface enables staff to manage the circulation and cataloging in a way that benefits patrons with the least amount of training is imperative.
 - b. Patrons using the system should have seamless experiences and get the most out of the system.
2. Comply with accessibility standards
 - a. Using the WCAG 2.1 guidelines the system must ensure it meets these standards for patrons with disabilities.
 - b. This can include features such as closed captioning for instructional videos, color contrasting settings, and multilingual support being available to patrons.
3. Mobile and remote access
 - a. Allowing patrons to access their accounts, materials, and know what the library can offer from their mobile devices.
 - b. Allowing patrons to access databases and digital resources ensures equity of access for all patrons.

4. Customization and Personalization

- a. The ability to create personalized accounts, save searches, and receive recommendations based on their history ensure patrons a better and more personal experience.
- b. This is beneficial for the staff as well if there is role-based access that aligns with different job responsibilities.

Prioritizing usability and access, SPL ensures its system serves a diverse and wide user base which directly aligns with their mission of providing equitable library services.

Cost of the System (Ashley)

Budgeting considerations unfortunately play one of the biggest impacting roles at any library. SPL must assess the total cost of ownership for the selected system ensuring that it fits within both the initial and long-term expense plans of the library.

1. Initial cost analysis

- a. Hardware requirements can be a factor in this if the new system requires additional servers or upgraded workstations or infrastructure.
- b. Licenses and fees can play a role with certain systems.
- c. If customizations are needed, then some vendors may charge extra.

2. Ongoing cost analysis

- a. The system may need maintenance or require support fees in the future.
- b. Subscription fees may be required.
- c. Staff may require training as you hire more staff or the system makes changes.

3. Cost-benefit analysis

- a. The SPL must weigh the long-term sustainability of this system and how it will affect budget constraints.

Support and Vendor Reputation (Ashley)

With any ILS, the success of implementation can rely on the vendor's credibility, long-term stability and its support structure. The SPL will need to evaluate the vendor based on the following criteria:

1. The vendor's track record
 - a. Assessing customer testimonials and industry reports will give a greater idea of the vendor and the system.
 - b. A well-established vendor reduces risk with the product becoming discontinued or having bad long-term support.
2. The vendor's customer support and training
 - a. A vendor should have technical support 24/7.
 - b. Training programs for the staff should be available in many different forms.
3. The customization and the ability of the system to integrate with the current situation
 - a. The ILS should be able to work with the current systems the library is using.
 - b. Customization should not lead to unnecessary or excessive costs.
4. The community and professional feedback
 - a. Consulting other professionals will allow SPL to know how the system will work with the staff.
 - b. Looking at forums and other reviews of the system could allow SPL to see if there are any significant flaws that others have noticed.

Careful research will allow SPL to know without a doubt if this system is the right fit for their library. Evaluating these factors will allow them to find a system that is reliable, cost-efficient, and long lasting for both the library staff and patrons.

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